SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Scenario-Based Questions

Course Code:	**DSA0405**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO4 – Scenario-Based Questions**	**Assessment:**	**Assessment Tool 1– Scenario-Based Questions**
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO4 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
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Section / Batch:	2024- CSE(AI)	**Date:**	20-08-26

Code of Conduct
I __S.Sirivally / 192472371__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: __Sirivally.S___

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	**25**	**Thorough understanding; clearly identifies all key issues**	**Good understanding with most issues identified**	**Basic understanding; some issues missed**	**Poor understanding; problem not identified correctly**	
Application of Concepts	**25**	**Effectively applies relevant concepts to analyze the case deeply**	**Applies appropriate concepts with minor gaps**	**Limited application of concepts**	**Fails to apply relevant concepts**	
Analysis	**20**	**In-depth analysis with strong reasoning and insights**	**Good analysis with logical reasoning**	**Basic analysis with limited depth**	**Weak or no analysis; lacks reasoning**	

Solution Design	**20**	**Innovative, feasible solutions with strong justification**	**Logical solutions with adequate justification**	**Basic solutions with weak justification**	**Incorrect or no solution provided**	
Clarity	**10**	**Clear, well-structured, and easy to understand**	**Organized and understandable presentation**	**Limited clarity and structure**	**Poor clarity; difficult to understand**	

QUESTION 1

A software company claims that its new application has an average response time of less than 2 seconds. A hypothesis test gives a p-value of 0.004. At $\alpha = 0.01$, determine whether there is sufficient evidence to support the company's claim.

Solution:

Population: All response times of users of the new application. This includes response times from the broader user population to which the company's claim is intended to apply.

Sample: The observations used to perform the hypothesis test. The sample size, individual response times, sample mean, and sample standard deviation are not provided.

Parameter of interest: Population mean response time, μ, measured in seconds.

Point estimate: The sample mean response time, $\bar{x}$, is the natural point estimate of μ. Its numerical value is not provided, so an exact numerical point estimate cannot be calculated.

Statistical claim: The company claims that the population mean response time is less than 2 seconds. Because the claim is directional, a one-sided lower-tail hypothesis test is appropriate.

Null hypothesis (H_0): $H_0: \mu \geq 2$ seconds.

Alternative hypothesis (H_1): $H_1: \mu < 2$ seconds.

Significance level: $\alpha = 0.01$, meaning the decision rule allows a 1% significance level for rejecting H_0.

Given p-value: $p = 0.004$.

Test statistic: The exact test statistic cannot be calculated from the information supplied. For a one-sample mean test, the calculation would normally require the sample mean, the hypothesized mean, the sample standard deviation or population standard deviation as appropriate, and the sample size. Since those values are absent, only the p-value-based decision can be completed.

Calculation:

Decision rule: Reject H_0 if p-value $< \alpha$.

Substitution: $0.004 < 0.01$.

Therefore: Reject H_0.

p-value interpretation: If the null hypothesis were true, the probability of obtaining evidence at least as strong as the observed evidence in the direction of a mean response time below 2 seconds would be 0.004, or 0.4%. Because this probability is very small, the observed result would be unusual under H_0.

Decision: Since $p = 0.004$ is less than $\alpha = 0.01$, reject H_0.

Conclusion: There is sufficient statistical evidence at the 1% significance level to support the company's claim that the new application has an average response time of less than 2 seconds.

Scenario-based interpretation: From a software-performance perspective, the result supports the claim that the application is meeting a sub-two-second average response-time target for the population represented by the study. The result is statistically strong because the p-value is smaller than the strict 1% significance threshold. However, the hypothesis test alone does not describe variability among individual users, peak-load performance, or the percentage of users experiencing slower responses.

Practical recommendation: The company can use the result as evidence supporting the response-time claim, but it should continue monitoring response times under different network conditions, device types, traffic levels, and peak workloads. These operational checks are important because an average below two seconds does not guarantee that every user experiences a response below two seconds.

QUESTION 2

A digital marketing team launches a new advertisement and wants to know whether it increases the click-through rate. Before the campaign, the rate was 6%. After the campaign, a hypothesis test produces a p-value of 0.12. What should the data science team conclude at the 5% significance level?

Solution:

Population: All users who could be exposed to the new advertisement and for whom a click-through outcome could be observed.

Sample: Users included in the post-campaign analysis. The sample size and number of clicks are not supplied.

Parameter of interest: The population click-through rate after the campaign, p.

Point estimate: The sample click-through rate, $\hat{p}$, is the point estimate of p. Its numerical value cannot be calculated because the number of users and number of clicks are not provided.

Statistical claim: The marketing team wants to determine whether the new advertisement increases the click-through rate above the previous 6% benchmark.

Null hypothesis (H_0): H_0: $p \leq 0.06$.

Alternative hypothesis (H_1): H_1: $p > 0.06$.

Significance level: $\alpha = 0.05$.

Given p-value: $p = 0.12$.

Test statistic: The exact test statistic cannot be calculated because the sample size and observed sample click-through rate are not supplied. A one-proportion test would normally use these quantities. The supplied p-value is sufficient to make the hypothesis-test decision.

Calculation:

Decision rule: Reject H_0 if p-value $< \alpha$.

Substitution: $0.12 < 0.05$ is false.

Equivalent comparison: $0.12 > 0.05$.

Therefore: Fail to reject H_0.

p-value interpretation: A p-value of 0.12 means that, if the null hypothesis were true, there would be a 12% probability of obtaining evidence at least as strong as the observed evidence in favor of a click-through rate above 6%. Since 12% is larger than the chosen 5% threshold, the evidence is not strong enough to establish an increase.

Decision: Since $p = 0.12 > \alpha = 0.05$, fail to reject H_0.

Conclusion: There is not sufficient statistical evidence at the 5% significance level to conclude that the new advertisement increased the click-through rate above 6%.

Important statistical interpretation: Failing to reject H_0 does not mean that the advertisement definitely did not increase the click-through rate. It means that the available evidence is insufficient to demonstrate an increase at the selected significance level.

Scenario-based interpretation: For the digital marketing team, the observed campaign result should not be described as a statistically proven improvement over the 6% benchmark. The p-value of 0.12 indicates that the evidence for an increase is relatively weak under the selected testing criterion.

Practical recommendation: The team should avoid making a strong performance claim based solely on this test. It may collect additional observations, evaluate audience segments, test alternative advertisements, and examine conversion and revenue metrics. If a larger or more representative sample provides stronger evidence, the team can reassess the campaign decision.

QUESTION 3

A machine-learning team develops a new classification model. The old model has an accuracy of 82%, while the new model achieves 85%. A statistical test gives a p-value of 0.01. Explain what the p-value means and whether the team should conclude that the new model performs significantly better.

Solution:

Population: The broader set of observations on which the classification models may be used. The population should represent the types of future observations for which model performance matters.

Sample: The test observations used to compare the old and new models.

Parameter of interest: The difference between the population accuracies of the new and old models, represented conceptually as $\mu_{new} - \mu_{old}$ when accuracy is treated as a population performance measure.

Point estimate: The observed difference in accuracy is the point estimate of the improvement.

Calculation:

Observed accuracy of new model = 85%.

Observed accuracy of old model = 82%.

Estimated improvement = 85% − 82% = 3 percentage points.

Interpretation of point estimate: The new model is observed to be 3 percentage points more accurate than the old model on the supplied test results. This is an observed difference and should not automatically be treated as the exact improvement in population performance.

Null hypothesis (H_0): H_0: The new model does not have higher population accuracy than the old model.

Alternative hypothesis (H_1): H_1: The new model has higher population accuracy than the old model.

Significance level: Assume $\alpha = 0.05$ unless otherwise stated.

Given p-value: $p = 0.01$.

Test statistic: The exact test statistic cannot be calculated from the supplied information because the test sample size, paired or independent comparison structure, and other quantities needed for the selected statistical test are not provided.

Meaning of p-value: Under the null hypothesis that the new model does not have higher population accuracy, a p-value of 0.01 indicates a 1% probability of obtaining evidence at least as strong as the observed evidence in favor of the new model being better. The small p-value provides strong evidence against H_0.

Calculation:

Decision rule: Reject H_0 if p-value $< \alpha$.

Substitution: $0.01 < 0.05$.

Therefore: Reject H_0.

Decision: Since $p = 0.01 < \alpha = 0.05$, reject H_0.

Conclusion: There is statistically significant evidence at the 5% significance level that the new classification model performs better than the old model in terms of population accuracy.

Scenario-based interpretation: The observed accuracy improvement is 3 percentage points, and the p-value indicates that such evidence would be relatively unlikely if the new model did not actually have higher population accuracy. Therefore, the team has statistical evidence supporting the improvement.

Practical significance: Statistical significance does not automatically mean that the improvement is practically important. A 3-percentage-point increase may be valuable in one application but less important in another. The team should consider the number and importance of errors, computational cost, prediction latency, memory requirements, robustness, fairness or stability across data groups where relevant, retraining cost, and business impact.

Practical recommendation: The team should consider the new model for further validation or deployment, but should not base the decision on p-value alone. Before full production use, it should verify performance on representative and preferably unseen data and compare the operational benefits of the 3-percentage-point improvement with additional resource requirements.

QUESTION 4

A factory claims that its defect rate is 2%. After randomly inspecting products, a 95% confidence interval for the defect rate is found to be (2.4%, 3.6%). At the same time, a hypothesis test gives a p-value of 0.01. Interpret both results and explain whether the company's claim should be rejected.

Solution:

Population: All products manufactured by the factory.

Sample: The randomly inspected products.

Parameter of interest: Population defect rate, p.

Point estimate: The exact sample defect rate is not provided. Therefore, the exact numerical point estimate cannot be calculated from the supplied information.

Null hypothesis (H_0): H_0: $p = 0.02$.

Alternative hypothesis (H_1): H_1: $p \neq 0.02$.

Confidence level: 95%.

Confidence interval: (2.4%, 3.6%).

Confidence interval interpretation: The interval gives a plausible range for the population defect rate based on the sample and the stated 95% confidence level. In practical terms, the analysis places the population defect rate between 2.4% and 3.6% according to the stated interval.

Comparison with the claimed rate: The company's claimed defect rate is 2%. The entire confidence interval lies above 2%, because its lower endpoint is 2.4%. Therefore, 2% is outside the confidence interval.

Calculation:

Lower confidence-limit comparison: $2.4\% > 2.0\%$.

Upper confidence-limit comparison: $3.6\% > 2.0\%$.

Therefore, every value in the interval is greater than the claimed 2% rate.

Given p-value: $p = 0.01$.

Significance level: $\alpha = 0.05$.

Test statistic: The exact test statistic cannot be calculated because the sample size and number of defective products are not provided. The supplied p-value and confidence interval are sufficient for the requested interpretation.

p-value interpretation: If the true defect rate were 2%, a p-value of 0.01 means that there would be a 1% probability of obtaining evidence at least as inconsistent with the 2% claim as the evidence observed, in the direction covered by the two-sided test. This is small enough to provide evidence against H_0.

Calculation:

Decision rule: Reject H_0 if p-value $< \alpha$.

Substitution: $0.01 < 0.05$.

Therefore: Reject H_0.

Decision: Reject H_0 at the 5% significance level.

Conclusion: There is sufficient statistical evidence to reject the claim that the factory's population defect rate is 2%. Both the p-value and the confidence interval point to the same conclusion.

Why the two methods agree: The confidence interval excludes the hypothesized value of 2%, while the p-value is below 0.05. These two results are consistent for a conventional two-sided test at the 5% significance level.

Scenario-based interpretation: The manufacturing evidence suggests that the factory's actual defect rate is higher than the claimed 2% level. This is important for quality control because even a difference of a few percentage points can represent a substantial number of defective units when production volume is high.

Practical recommendation: The factory should investigate production stages, raw materials, machine calibration, inspection procedures, and recurring defect patterns. Management should also monitor the defect rate after corrective actions to determine whether the process has returned to an acceptable quality level.

QUESTION 20

An online shopping company wants to determine whether a new user-interface design increases the average amount spent per customer. The data scientist calculates a 95% confidence interval for the difference in average spending as (₹120, ₹480) and obtains a p-value of 0.002. Interpret the confidence interval and p-value, and provide a business recommendation.

Solution:

Population: Customers represented by the study and the wider customer population to which the result is intended to apply.

Sample: Customers included in the comparison between the old and new interface.

Parameter of interest: The difference in population mean spending between customers using the new interface and customers using the old interface, represented as $\mu_{new} - \mu_{old}$.

Point estimate: The exact observed difference is not provided. The confidence interval indicates a plausible positive increase between ₹120 and ₹480 per customer.

Null hypothesis (H_0): H_0: $\mu_{new} - \mu_{old} \leq 0$.

Alternative hypothesis (H_1): H_1: $\mu_{new} - \mu_{old} > 0$.

Significance level: $\alpha = 0.05$.

Confidence level: 95%.

Confidence interval: (₹120, ₹480).

Confidence interval interpretation: The 95% confidence interval indicates that the plausible increase in population average spending associated with the new interface is between ₹120 and ₹480 per customer, based on the study.

Comparison with zero effect: The entire confidence interval is above ₹0. Therefore, the interval contains only positive differences and does not include zero.

Calculation:

Lower limit = ₹120.

Upper limit = ₹480.

Both limits are greater than ₹0.

Therefore, the interval supports a positive difference in average spending.

Given p-value: $p = 0.002$.

Test statistic: The exact test statistic cannot be calculated from the supplied information because the sample sizes, sample means, and variability measures are not given. The p-value is sufficient for the hypothesis-test decision.

p-value interpretation: If the null hypothesis of no positive increase were true, a p-value of 0.002 would mean a probability of only 0.2% of obtaining evidence at least as strong as the observed evidence in favor of increased average spending. This is very small and provides strong evidence against H_0.

Calculation:

Decision rule: Reject H_0 if p-value $< \alpha$.

Substitution: $0.002 < 0.05$.

Therefore: Reject H_0.

Decision: Reject H_0 at the 5% significance level.

Conclusion: There is statistically significant evidence that the new user-interface design increases average spending per customer.

Scenario-based interpretation: The confidence interval suggests that the increase may be economically meaningful, with a plausible range from ₹120 to ₹480 per customer. The p-value indicates strong statistical evidence that the increase is not simply attributable to random sampling variation under the null hypothesis.

Business recommendation: The company should consider implementing or expanding the new interface, preferably through a controlled rollout and continued measurement. The company should evaluate implementation cost, customer experience, customer retention, long-term performance, and whether the observed increase persists over time. It should also monitor whether higher spending is accompanied by sustainable customer value rather than unintended effects such as increased returns or reduced satisfaction.

Additional business consideration: The confidence interval is especially useful for decision-making because it provides a range of plausible effect sizes rather than only a significance decision. If the implementation cost is small relative to the expected additional spending and the effect remains stable, the business case for wider deployment becomes stronger. If implementation is expensive, the company should compare those costs against the lower end of the estimated increase.